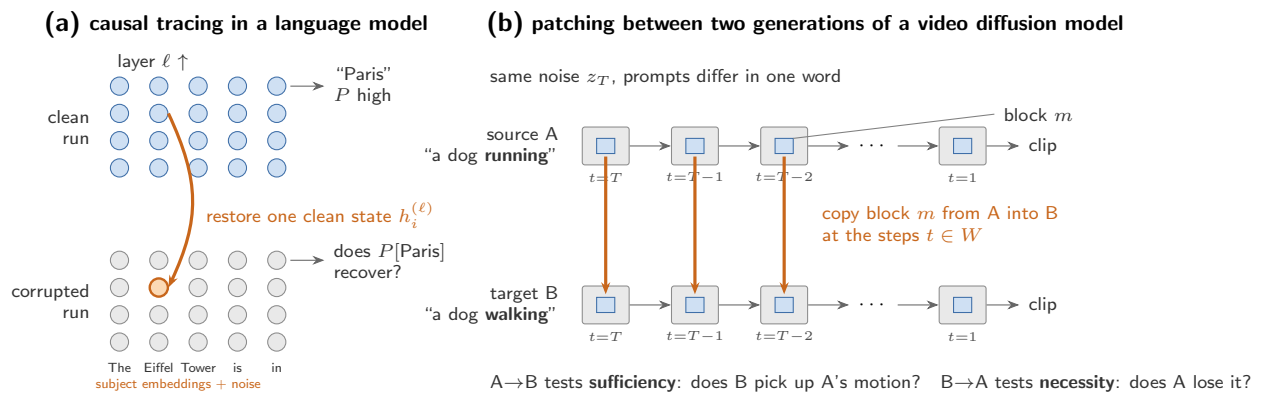

(a) causal tracing in a language model
layer ℓ ↑
"Paris"
P high
clean
run
restore one clean state $h_i^{(\ell)}$
does P[Paris]
recover?
corrupted
run
The
Eiffel
Tower
is
in
subject embeddings + noise
(b) patching between two generations of a video diffusion model
same noise z_T, prompts differ in one word
source A
"a dog running"
block m
clip
$t=T$
$t=T-1$
$t=T-2$
$t=1$
copy block m from A into B
at the steps $t \in W$
target B
"a dog walking"
clip
A→B tests sufficiency: does B pick up A's motion?
B→A tests necessity: does A lose it?